

Online Cake Shop – Project Setup for the Complete Business Analysis Training Portfolio

1. Purpose

The **Online Cake Shop** project has been selected as the primary case study for this Business Analysis training program. Instead of creating isolated assignments, the project will evolve throughout the course and gradually produce every major Business Analysis artifact expected in a real software development project.

The project begins with requirement analysis and Agile planning in Assignment 2 and continues through business modeling, process analysis, UML design, requirement specifications, testing, documentation, and finally an AI-enabled capstone project.

This approach demonstrates how a Business Analyst contributes throughout the entire Software Development Life Cycle (SDLC).

2. Why the Online Cake Shop?

The Online Cake Shop provides an ideal learning project because it includes:

- Customer registration and authentication
- Product catalogue management
- Shopping cart
- Online ordering
- Payment processing
- Delivery scheduling
- Order tracking
- Customer account management
- Administrative functions
- Inventory management
- Reporting

These features cover almost every Business Analysis technique taught during the course.

3. Relationship to Other Course Projects

Although the Online Cake Shop is the primary project, several assignments introduce additional projects to expose students to different business domains.

Project	Purpose
Online Cake Shop	End-to-end SDLC, Agile, UML, Requirements, Testing, Capstone

Project	Purpose
Job Recruitment Portal	Requirement Elicitation, BPMN, User Stories, Scope, Change Requests
Online Repair Shop	Database Design, SQL, BI Dashboard

The skills learned from these projects are complementary. The same BA techniques are applied across different industries, demonstrating that Business Analysis is domain-independent.

4. Project Vision

Develop a modern online bakery platform that enables customers to browse products, customize cakes, place secure online orders, make payments, and track deliveries through both web and mobile applications while providing administrators with tools to manage inventory, orders, customers, and business operations.

5. Business Objectives

The system should:

- Increase online sales.
 - Improve customer experience.
 - Reduce manual order processing.
 - Provide real-time order tracking.
 - Enable secure digital payments.
 - Improve inventory management.
 - Generate business analytics and reports.
 - Support future AI-powered recommendations.
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6. Project Scope

In Scope

- User Registration
- Login
- Customer Profile
- Product Catalogue
- Search and Filters
- Cake Customization
- Shopping Cart
- Checkout
- Online Payment
- Order Confirmation
- Order Tracking
- Delivery Scheduling

- Customer Reviews
- Promotions
- Admin Dashboard
- Product Management
- Inventory Management
- Reporting

Out of Scope

- Physical store management
 - Manufacturing automation
 - Supplier procurement
 - Warehouse robotics
 - Accounting system
 - HR management
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7. Technology Stack

Frontend

- Responsive Web Application
- Android Application
- iOS Application

Possible Technologies

- React
 - Angular
 - Flutter
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Backend

- REST APIs
- Authentication Service
- Product Service
- Order Service
- Payment Service
- Notification Service

Possible Technologies

- Java Spring Boot
 - Node.js
 - Python FastAPI
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Database

Relational Database

Possible Options

- PostgreSQL
- MySQL

Stores

- Customers
 - Products
 - Orders
 - Payments
 - Inventory
 - Delivery Information
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Third-Party Integrations

- Razorpay / Stripe
 - Google Maps
 - Email Notifications
 - SMS Notifications
 - Push Notifications
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8. Open Source Tools Used Throughout the Course

Tool	Purpose
OpenProject / Taiga	Agile Backlog, Sprint Planning
LibreOffice Writer	BRD, FRD, SRS, Reports
LibreOffice Calc	RICE, MoSCoW, RTM
LibreOffice Impress	Presentations
Draw.io	Flowcharts, UML, Context Diagrams
PlantUML	UML as Code
BPMN.io / Camunda	BPMN Models
Excalidraw	Brainstorming
Penpot	Wireframes

Tool	Purpose
PostgreSQL	Database
DBeaver	ERD
Metabase	BI Dashboard
GitHub	Version Control
Ollama	AI-assisted BA activities

9. SDLC Progression Across the Course

Instead of producing isolated documents, every assignment contributes to the same project.

Assignment 2

- SDLC Mapping
- Requirement Summary
- Product Backlog
- Agile Board
- Prioritization

Assignment 3

- Stakeholder Analysis
- Elicitation Plan
- Scope & Vision
- Context Diagram

Assignment 4

- Business Process
- Use Cases
- Context Diagram

Assignment 5

- BPMN
- User Stories
- INVEST
- Gherkin
- Story Estimation

Assignment 6

- Functional Requirements
- Software Requirement Specification
- Requirement Traceability Matrix

Assignment 7

- Activity Diagram
- Sequence Diagram
- Class Diagram

Assignment 8

- Professional Documentation
- Presentations

Assignment 9

- Wireframes
- UI/UX
- Accessibility

Assignment 10

- ER Diagram
- Database Design
- SQL
- Business Intelligence Dashboard

Assignment 11

- Project Planning
- Change Requests
- User Manual
- Estimation

Assignment 12

- Test Cases
- BDD
- API Testing
- Automation
- Security Testing

Capstone

The final project combines all previous deliverables into one complete Business Analysis portfolio.

10. AI Integration Throughout the Course

Artificial Intelligence is used as a productivity assistant rather than a replacement for Business Analysis.

Examples include:

- Drafting user stories

- Generating interview questions
- Identifying missing requirements
- Reviewing SRS documents
- Creating UML drafts
- Generating SQL queries
- Drafting test cases
- Improving documentation

Every AI-generated output will be reviewed, corrected, and validated by the Business Analyst before inclusion in the project.

11. Future AI Feature (Capstone)

The Online Cake Shop will be enhanced with an AI-powered recommendation engine capable of:

- Recommending cakes based on customer preferences.
- Suggesting products based on previous purchases.
- Supporting natural language ordering (e.g., "I need a chocolate birthday cake for 15 people under \$50.").
- Predicting popular products during festivals and holidays.

This AI feature provides an opportunity to learn AI requirements engineering, explainability, governance, privacy, and human oversight as part of the capstone project.

12. Expected Portfolio Structure

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ba-training-portfolio/  
|  
├─ assignment-01/  
├─ assignment-02/  
├─ assignment-03/  
├─ assignment-04/  
├─ assignment-05/  
├─ assignment-06/  
├─ assignment-07/  
├─ assignment-08/  
├─ assignment-09/  
├─ assignment-10/  
├─ assignment-11/  
├─ assignment-12/  
|  
├─ capstone/  
|  
├─ diagrams/  
├─ templates/  
└─ README.md
```

```
|
└─ Online_Cake_Shop_Project/
    └─ Vision
    └─ Requirements
    └─ UML
    └─ BPMN
    └─ Database
    └─ Wireframes
    └─ Testing
    └─ AI Feature
    └─ Documentation
```

13. Conclusion

The Online Cake Shop serves as the central project that demonstrates the complete Business Analysis lifecycle. Rather than creating disconnected assignment submissions, each assignment extends the same project with additional artifacts, allowing students to build a professional portfolio that mirrors real-world software development. Complementary projects such as the Job Recruitment Portal and Online Repair Shop reinforce the same BA techniques across different domains, preparing learners to apply Business Analysis practices in any industry.